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PARTIAL EUROPEAN SEARCH REPORT

under Rule 64, paragraph 1 of the European Patent Convention

Application Number EP 22 16 5219

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
A	LINHART B. ET AL.: "Molecular approaches for diagnosis, therapy and prevention of cow's milk allergy", NUTRIENTS, vol. 11, no. 7, 1492, 29 June 2019 (2019-06-29), pages 1-24, XP055949535, * the whole document * * figure 1 * * table 1 *	1-12	INV. G01N33/68 A61P37/08 C07K14/47 C07K16/00
A	HOCHWALLNER H. ET AL.: "Cow's milk allergy: From allergens to new forms of diagnosis, therapy and prevention", METHODS, vol. 66, no. 1, March 2014 (2014-03), pages 22-33, XP055951224, * the whole document * * table 1 * * figures 2,3 *	1-12	

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TECHNICAL FIELDS SEARCHED (IPC)

G01N
C07K
A61P

LACK OF UNITY OF INVENTION

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

see sheet B

The present partial European search report has been drawn up for those parts of the European patent application which relate to the invention first mentioned in the claims.

Place of search	Date of completion of the search	Examiner
Munich	12 August 2022	Giry, Murielle

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

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EPO FORM 1503 03.82 (P04E27)

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PARTIAL EUROPEAN SEARCH REPORT

Application Number
EP 22 16 5219

DOCUMENTS CONSIDERED TO BE RELEVANT			CLASSIFICATION OF THE APPLICATION (IPC)
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	
X	WO 2015/048339 A2 (PRONUTRIA, INC. [US]) 2 April 2015 (2015-04-02) * abstract * * claims 1-82 * * SEQ ID NO: 1043 * -----	1,2,4-6, 9,11	
A	WO 2010/043724 A1 (MEAD JOHNSON & COMPANY [US]) 22 April 2010 (2010-04-22) * the whole document * -----	1-12	
			TECHNICAL FIELDS SEARCHED (IPC)

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EPO FORM 1503 03.82 (P04C10)

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LACK OF UNITY OF INVENTION
SHEET B

Application Number
EP 22 16 5219

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

1. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E1) and (E2) belonging to immunoglobulin M heavy chain secretory form

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2. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E3), (E4), (E5), (E6), (E7) and (E8) belonging to ?-1 metal binding globulin

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3. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E9) and (E10) belonging to lactoperoxidase

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4. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E11) and (E12) belonging to polymeric immunoglobulin receptor

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5. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E13) belonging to C3-beta-c

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6. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E14) and (E15) belonging to Ig heavy chain precursor (B/MT.4A.17. H5. A5)

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7. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E16) and (E17) belonging to perilipin

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8. claims: 1-12(partially)

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E18) and (E19) belonging to zinc-alpha-2-glycoprotein

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9. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E20) and (E21) belonging to glycoprotein 2

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10. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E22) belonging to lactadherin

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11. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E23) belonging to immunoglobulin light chain, lambda gene cluster

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12. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E24) belonging to intracellular cholesterol transporter 2

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13. claims: 1-12(partially)

compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E25) belonging to apolipoprotein A-IV

**The present application discloses the identification of antigens responsible of allergy and their use for diagnosis purposes.
In the light of the prior art documents, the problem to be solved by the present application can be seen in providing alternative compositions and methods for diagnosing an allergy.
The polypeptides defined by (E1) to (E25) in claim 1 provide 13 solutions to the problem posed.**

The common matter which can be identified as linking together the various polypeptides is that as they are identified as being milk derived antigens.

LACK OF UNITY OF INVENTION
SHEET B

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The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

However, milk-derived antigens are known from the prior art. D1 discloses the identification of milk allergens, including among others caseins, alpha-lactalbumin, beta-lactoglobulin and lactoferrin and recombinant epitopes thereof (Figure 1, Table 1) as well as hypoallergenic milk formulas. The alternative allergens of claim 1 only have in common that they are milk allergens. As this common feature has already been disclosed in the prior art it cannot be regarded as the special technical feature providing unity among said alternatives.
Hence, the above identified common general concept is not new and thus cannot be the single general inventive concept as required by Art. 82 EPC. The present application is therefore considered not to fulfil the requirements of unity as laid down in Art. 82 EPC. Therefore the claimed antigens according to the identified 13 groups of invention represent a distinct solution to the technical problem of providing further compositions and methods for diagnosing an allergy.

Moreover, the mere fact that the proteins are derived from the same source, namely milk is not sufficient to meet the criteria for unity of invention.

Due to the essential differences in the antigens contemplated as markers, which do not share a common structure, property or activity, no other technical features could be identified that form a technical relationship among each of the separate inventions claimed and which could be considered as corresponding special technical features in the sense of Rule 44(1) EPC.

The search has been limited to the invention first mentioned in the claims.
As the searches for inventions 2-13 represent a major extra search burden, the applicant is invited to pay 12 additional search fees.

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 22 16 5219

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

12-08-2022

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2015048339 A2	02-04-2015	NONE	
WO 2010043724 A1	22-04-2010	BR PI0919789 A2	03-05-2016
		CA 2740255 A1	22-04-2010
		CN 102187226 A	14-09-2011
		CO 6351822 A2	20-12-2011
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		EP 2347264 A1	27-07-2011
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		NO 342293 B1	30-04-2018
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		PL 2347264 T3	30-06-2014
		RU 2011119613 A	27-11-2012
		TW 201030341 A	16-08-2010
		US 2010143262 A1	10-06-2010
		WO 2010043724 A1	22-04-2010